

# MODULE - 5

## Bivariate Data.

Data based on the characteristics of two variables are called bivariate data. The variables are usually denoted by  $x$  and  $y$ .

## Correlation

It is the degree of relationship between two variables.

Two Variables are said to be Correlated if the change in One variable results in a corresponding change in the other variable.

eg:- When the price of a Commodity increases the supply for that Commodity decreases.

On the other hand, If price decreases, supply increases. That means, both variables move in the opposite direction. Hence price and supply are correlated.

The word Correlation implies, Cause and effect relationships.

The increase in Cigarette Smoking and increasing in Cancer patients, proves that one Cause the other.

## Different types of Correlation.

### 1) Positive and negative Correlation

When the value of two Variables move in the same direction, Correlation is said to be positive.

ie. increase in One variable, results in the increase in other variable or decrease in One variable results in decrease in other variable, we say that the two variables are positively Correlated.

eg:- Consider the bivariate data  $x$  and  $y$ .

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | 30 | 35 | 42 | 43 | 50 |
| $y$ | 12 | 15 | 17 | 20 | 25 |

} positive Correlation

Both variables move in the same direction.

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | 20 | 25 | 30 | 49 | 50 |
| $y$ | 40 | 35 | 25 | 20 | 10 |

} negative Correlation

Here  $x$  increases and  $y$  decreases.



ie  $x$  and  $y$  move in the opposite direction.

$\therefore x$  and  $y$  are negatively correlated.

## 2) Linear and Non-Linear Correlation

Correlation may be linear or non-linear. When the amount of change in one variable leads to a constant ratio of change in the other variable, correlation is said to be linear.

The graph of a linear correlation is always a straight line.

eg:- Consider the bivariate data  $x$  &  $y$ .

$x : 30 \quad 35 \quad 40 \quad 45 \quad 50$

$y : 12 \quad 15 \quad 18 \quad 21 \quad 24$

Correlation is said to be non-linear when the amount of change in one variable is not a constant ratio to the amount of change in the other variable.

The graph of a non-linear correlation is a curve and  $\therefore$  is it also called Curvi-

## Linear Correlation.

eg:- Consider the bivariate data

$x$  : 20 25 30 35 40

$y$  : 12 15 25 37 50

$$\frac{5}{3}$$

$$\frac{5}{10}$$

$$\frac{5}{12}$$

$$\frac{5}{13}$$

### 3) Perfect and Imperfect Correlation.

The Correlation between two Variables is usually denoted by  $R$ . The value of  $r$  lies between  $-1$  &  $+1$ , including  $-1$  and  $+1$ .

ie.  $-1 \leq r \leq 1$

If the value of Correlation between the two variables  $x$  and  $y$  is  $1$  or  $-1$ , we say that there is perfect Correlation. If the value of  $r$  is in between  $-1$  &  $+1$ , we say that the Correlation is imperfect.

### Measures of Correlation.

#### 1. Scatter Diagrams.

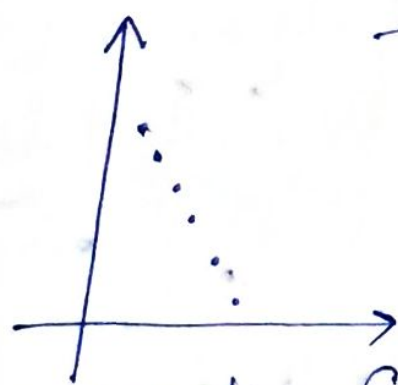
This is a graphical method of Studying Correlation between two Variables.



One of the variable is plotted on the x-axis and the other variable is plotted on y-axis. Each pair of values  $(x, y)$  is plotted on the graph by means of dot(.) is called Scatter diagram, for a diagram of dots

→ If the points show some trend upward or downward, the two variables are said to be Correlated.

→ If the plotted dots do not show any trend, the 2 variables are not-Correlated.



negative-Correlation

from a given set of data, draw Scatter diagrams and Comment the relationship b/w 2 variables.

Set 1.

| x  | y  |
|----|----|
| 5  | 6  |
| 8  | 10 |
| 2  | 3  |
| 4  | 4  |
| 10 | 12 |

|   |   |
|---|---|
| 6 | 8 |
| 8 | 8 |

Set 2.

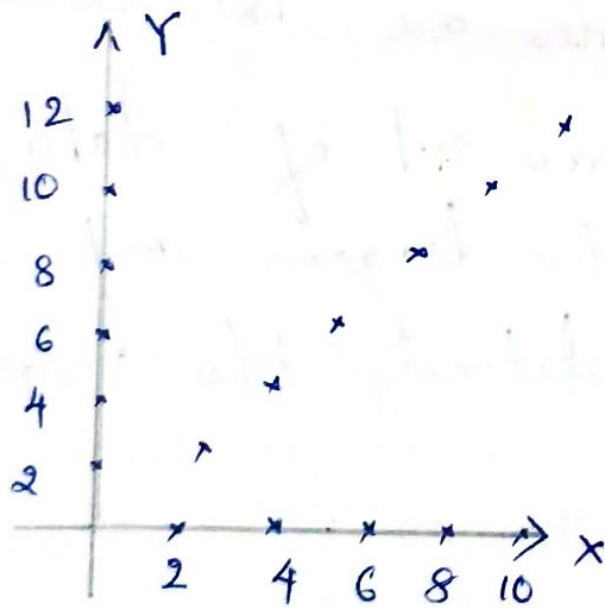
| x  | y  |
|----|----|
| 12 | 4  |
| 7  | 3  |
| 10 | 4  |
| 8  | 8  |
| 4  | 10 |

|   |   |
|---|---|
| 9 | 8 |
| 2 | 8 |

Set 3

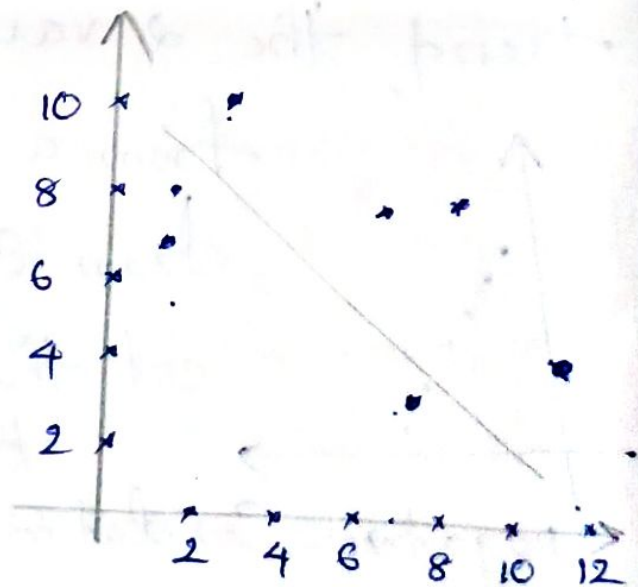
| x  | y  |
|----|----|
| 2  | 3  |
| 5  | 2  |
| 7  | 10 |
| 8  | 1  |
| 3  | 8  |
| 9  | 3  |
| 11 | 4  |

Set 1



Positive Correlation

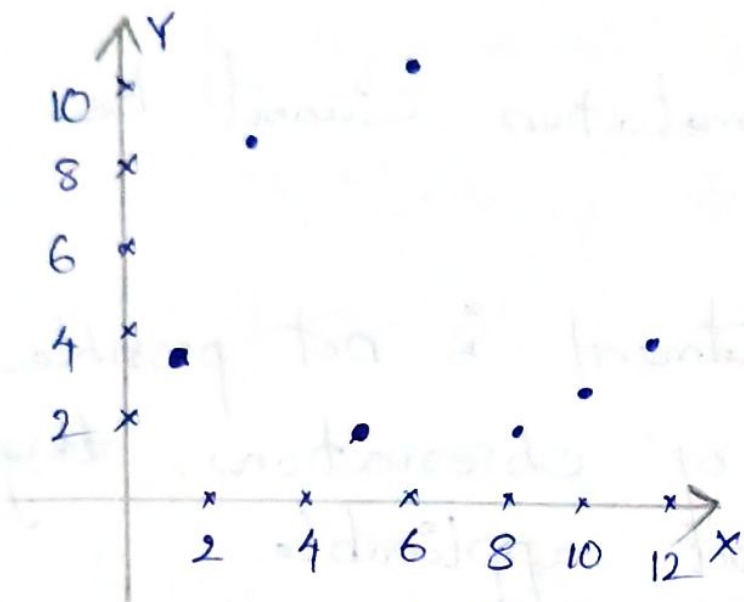
Set 2



Negative Correlation



Set 3.



No-Correlation.

From the 3 scatter diagrams, we can conclude that there is +ve correlation between  $x$  and  $y$  in set 1; there is -ve correlation b/w  $x$  and  $y$  in set 2 and there is no correlation b/w  $x$  and  $y$  in set 3.

Merits and demerits of Scatter diagram

Merits.

- it is easy to plot point
- it is simple to understand
- Extreme values do not affect it.
- abnormal values are easily detected in the data.

demerits

- $\Rightarrow$  degree of Correlation Cannot be estimated
- $\Rightarrow$  algebraic treatment is not possible
- $\Rightarrow$  when the no: of observations large this method is not applicable.

Karl Pearson's Coefficient of Correlation

Suppose we are given a bivariate data of  $x$  and  $y$ , then the Karl Pearson's Coefficient of Correlation denoted by  $r$ .

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}$$



2) Calculate Karl Pearson's Coefficient of Correlation for the following data.

$x : 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8$   
 $y : 4 \quad 5 \quad 6 \quad 12 \quad 9 \quad 5 \quad 4$

Ans:-

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 2   | 4   | 4     | 16    | 8    |
| 3   | 5   | 9     | 25    | 15   |
| 4   | 6   | 16    | 36    | 24   |
| 5   | 12  | 25    | 144   | 60   |
| 6   | 9   | 36    | 81    | 54   |
| 7   | 5   | 49    | 25    | 35   |
| 8   | 4   | 64    | 16    | 32   |

$$\Sigma x = 35 \quad \Sigma x^2 = 203 \quad \Sigma y^2 = 343$$

$$\Sigma y = 45 \quad \Sigma xy = 228$$

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{7 \times 228 - (35)(45)}{\sqrt{7 \times 203 - (35)^2} \sqrt{7 \times 343 - (45)^2}}$$

$$= \frac{1596 - 1575}{\sqrt{1421 - 1225} \sqrt{2401 - 2025}}$$

$$= \frac{21}{14 \times 19.39}$$

$$= \frac{21}{271.46} = 0.077$$

$$= 0.077$$

$$= 0.077$$

$$= 0.077$$

There is positive Correlation b/w two variables, and a very low degree of relationship b/w the variables.



? Calculate the Coefficient of Correlation for the following data

$x$  : 51      54      55      59      65      60      70

$y$  : 38      44      33      36      33      23      10

Ans: -

| $x$ | $y$ | $d_x = x - 59$ | $d_y = y - 33$ | $d_x^2$ | $d_y^2$ | $d_x d_y$ |
|-----|-----|----------------|----------------|---------|---------|-----------|
| 51  | 38  | -8             | 5              | 64      | 25      | -40       |
| 54  | 44  | -5             | 11             | 25      | 121     | -55       |
| 55  | 33  | -4             | 0              | 16      | 0       | 0         |
| 59  | 36  | 0              | 3              | 0       | 9       | 0         |
| 65  | 33  | 6              | 0              | 36      | 0       | 0         |
| 60  | 23  | 1              | -10            | 1       | 100     | -10       |
| 70  | 10  | 11             | -23            | 121     | 529     | -253      |
|     |     | 1              | -14            | 263     | 784     | -358      |

$$r = \frac{\sum d_x d_y - \frac{(\sum d_x)^2}{n}}{\sqrt{n \sum d_x^2 - \frac{(\sum d_x)^2}{n}} \sqrt{n \sum d_y^2 - \frac{(\sum d_y)^2}{n}}}$$

$$= \frac{7 \times -358 - 1(-14)}{\sqrt{7 \times 263 - 1^2} \sqrt{7 \times 784 - (-14)^2}}$$

$$\sqrt{7 \times 263 - 1^2} \sqrt{7 \times 784 - (-14)^2}$$

$$= -2506 + 14$$

$$\sqrt{1841-1} \sqrt{5488-196}$$

$$= \frac{-2492}{428 \times 72.7} = \frac{-2492}{3111.56} = \underline{\underline{-0.80}}$$

High degree of negative Correlation between  $x$  and  $y$ .

### Spearman's Rank Correlation.

Instead of giving the values of the variables, the ranks of observations are given, the Correlation Coefficient so obtained is called Rank Correlation Coefficient. It is denoted by  $\rho$  or  $\rho_{xy}$ .

When the ranks are distinct, Spearman developed a formula for rank Correlation which is given by  $\rho = 1 - \frac{6 \sum D^2}{n(n^2-1)}$



Where  $D$  = difference between ranks of  $x$  and  $y$ .

$n$  - no. of observations.

When the ranks are repeated, rank correlation is given by 
$$\rho = \frac{1}{n(n^2-1)} \left( 6 \sum D^2 + \frac{\sum (m^3 - m)}{12} \right)$$

Where  $D$  - difference b/w ranks of  $x$  &  $y$ .  
 $m$  - no. of times the rank repeated  
 $n$  - no. of observations.

?) Calculate Rank Correlation Coefficient between  $x$  and  $y$  for the following data

|                  | 1 | 2 | 3 | 4 |
|------------------|---|---|---|---|
| Ranks in maths : | 1 | 2 | 3 | 4 |

|                 |   |   |   |   |
|-----------------|---|---|---|---|
| Ranks in Stat : | 3 | 4 | 2 | 1 |
|-----------------|---|---|---|---|

| $x$ | $y$ | $D = x - y$ | $D^2$ |
|-----|-----|-------------|-------|
| 1   | 3   | -2          | 4     |
| 2   | 4   | -2          | 4     |
| 3   | 2   | 1           | 1     |
| 4   | 1   | 3           | 9     |

$$\sum D^2 = 18$$

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 18}{4 \times 15}$$

$$= 1 - \frac{108}{60} = 1 - 1.8 = \underline{\underline{-0.8}}$$

? Calculate Rank Correlation for the following data of ranks of 10 Competitors in a music Contest that 2 judges awarded.

Judge 1 : 5    4    2    6    7    10    9    1    8    3  
 Judge 2 : 4    1    5    10    8    9    7    6    3    2



| x  | y  | D  | D <sup>2</sup> |
|----|----|----|----------------|
| 5  | 4  | 1  | 1              |
| 4  | 1  | 3  | 9              |
| 2  | 5  | -3 | 9              |
| 6  | 10 | -4 | 16             |
| 7  | 8  | -1 | 1              |
| 10 | 9  | 1  | 1              |
| 9  | 7  | 2  | 4              |
| 1  | 6  | -5 | 25             |
| 8  | 3  | 5  | 25             |
| 3  | 2  | 1  | 1              |

$$\sum D^2 = 92$$

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 92}{10 \times 99}$$

$$= 1 - \frac{552}{990} = 1 - 0.551 = \underline{\underline{0.442}} = \underline{\underline{0.45}}$$

? Calculate the rank Coefficient for following data of weight of father and son.

Weight of dad : 57 54 55 59

65 60 70

Weight of Son : 38 44 33 36

34 23 10

| $x$ | $y$ | $R(x)$ | $R(y)$       | $D$          | $D^2$         |
|-----|-----|--------|--------------|--------------|---------------|
| 57  | 38  | 5      | 2            | 3            | 9             |
| 54  | 44  | 7      | 1            | 6            | 36            |
| 55  | 33  | 6      | <del>5</del> | <del>3</del> | <del>9</del>  |
| 59  | 36  | 4      | 3            | 1            | 1             |
| 65  | 34  | 2      | <del>4</del> | -2           | <del>16</del> |
| 60  | 23  | 3      | 6            | -3           | 9             |
| 70  | 10  | 1      | 7            | -6           | 36            |

$$r = 1 - \frac{6 \times 96}{7(7^2 - 1)}$$

$$= 1 - \frac{576}{336} = 1 - 1.71$$

$$= -0.71$$

$$\sum D^2 = 96$$



? Calculate Rank Correlation Coefficient for the following data on height of father & Son.

Height of father - 65 63 67 64 63 62  
70 64 68 65

height of Son - 68, 66 68 65 67,  
66, 68, 65, 71, 67

| $x$ | $y$ | Rank of $x$ | Rank of $y$ | $D$ | $D^2$               |
|-----|-----|-------------|-------------|-----|---------------------|
| 65  | 68  | 4.5         | 3           | 1.5 | 2.25                |
| 63  | 66  | 8.5         | 7.5         | 1   | 1                   |
| 67  | 68  | 3           | 3           | 0   | 0                   |
| 64  | 65  | 6.5         | 9.5         | -3  | 9                   |
| 63  | 67  | 8.5         | 5.5         | 3   | 9                   |
| 62  | 66  | 10          | 7.5         | 2.5 | 6.25                |
| 70  | 68  | 1           | 3           | -2  | 4                   |
| 64  | 65  | 6.5         | 9.5         | -3  | 9                   |
| 68  | 71  | 2           | 1           | 1   | 1                   |
| 65  | 67  | 4.5         | 5.5         | -1  | 1                   |
|     |     |             |             |     | $\Sigma D^2 = 42.5$ |

$$65 \text{ repeats twice } m^3 - m = 6 \quad (m=2)$$

$$63 \quad " \quad " \quad m^3 - m = 6$$

$$64 \quad " \quad " \quad m^3 - m = 6.$$

$$68 \text{ repeats thrice } m=3, m^3 - m = 24$$

$$66 \text{ repeats twice } m=2, m^3 - m = 6.$$

$$65 \quad " \quad " \quad m^3 - m = 6.$$

$$67 \quad " \quad " \quad m=2, m^3 - m = 6.$$

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$$\sum m^3 - m = 60.$$

$$\rho = 1 - 6 \left( \frac{\sum D^2 + \frac{\sum m^3 - m}{12}}{n(n^2 - 1)} \right)$$

$$= 1 - 6 \left( 42.5 + \frac{60}{12} \right) \frac{1}{10(10^2 - 1)}.$$

$$= 1 - \frac{6 \times 47.5}{990} = 1 - \frac{285}{990}$$

$$= 1 - 0.287 = 0.713$$



Calculate the rank Correlation Coefficient of the following data of marks of 5 students in an examination.

Marks in English :- 45 43 45 40 35

Marks in accountancy :- 50 25 50 20 34

| <u>Ans</u> - $x$ | $y$ | Rank of $x$ | Rank of $y$ | $D$ | $D^2$ |
|------------------|-----|-------------|-------------|-----|-------|
| 45               | 50  | 1.5         | 1.5         | 0   | 0     |
| 43               | 25  | 3           | 4           | -1  | 1     |
| 45               | 50  | 1.5         | 1.5         | 0   | 0     |
| 40               | 20  | 4           | 5           | -1  | 1     |
| 35               | 34  | 5           | 3           | 2   | 4     |

45 repeats twice  
 $m^3 - m = 6$

$$\sum D^2 = 6$$

50 repeats twice

$$m^3 - m = 6 \quad \sum m^3 - m = 12$$

$$\rho = 1 - \frac{6 \left( \sum D^2 + \frac{\sum m^3 - m}{12} \right)}{n(n^2 - 1)}$$

$$= 1 - \frac{6(6 + 1)}{120}$$

$$= 1 - \frac{42}{120} = 1 - 0.35$$

$$= \underline{\underline{0.65}}$$

Ques:-

The marks of 11 students in 2 subjects are given below.

x: 80 45 55 58 55 60 45 68 70  
45 85

y: 82 56 50 43 56 62 64 65  
70 64 90

Calculate the Coefficient of Correlation by method of ranks.



| $x$ | $y$ | Rank of<br>$x$ | Rank of<br>$y$ | $D$  | $D^2$ |
|-----|-----|----------------|----------------|------|-------|
| 80  | 82  | 2              | 2              | 0    | 0     |
| 45  | 56  | 10             | 8.5            | 1.5  | 2.25  |
| 55  | 50  | 7.5            | 10             | -2.5 | 6.25  |
| 58  | 43  | 6              | 11             | -5   | 25    |
| 55  | 56  | 7.5            | 8.5            | -1   | 1     |
| 60  | 62  | 5              | 7              | -2   | 4     |
| 45  | 64  | 10             | 5.5            | 4.5  | 20.25 |
| 68  | 65  | 4              | 4              | 0    | 0     |
| 70  | 70  | 3              | 3              | 0    | 0     |
| 45  | 64  | 10             | 5.5            | 4.5  | 20.25 |
| 85  | 90  | 1              | 1              | 0    | 0     |

$$\sum D^2 = 79$$

55 repeats twice  $m^3 - m = 6$

<sup>27-3</sup> 45 repeats thrice  $m^3 - m = 24$

56 " twice  $m^3 - m = 6$

64 " twice  $m^3 - m = 6$

$$\sum m^3 - m = 42$$

$$f = 1 - \frac{6 \left( \sum D^2 + \frac{\sum m^3 - m}{12} \right)}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \left( 79 + \frac{42}{12} \right)}{11 \times 120} = 1 - \frac{495}{1320} = 0.625$$

# Merits and demerits.

## Merits.

1. it is easy to calculate
2. Simple to understand
3. It can be applied in both quantitative & qualitative data

## A. demerits.

1. It is only an approximate measure because actual value is not used.
2. not applicable when  $n$  is large.
3. not Capable of further algebraic treatment.

✕

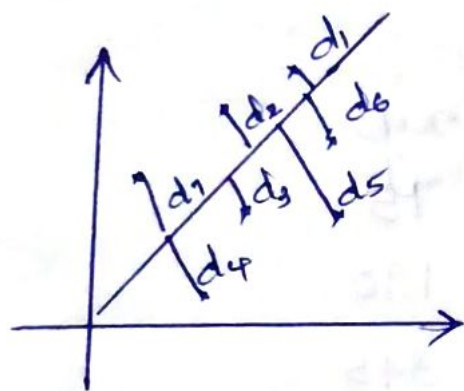
## Principle of least Squares.

Suppose we are given a bivariate data of  $x$  and  $y$  then draw a scatter diagram by taking variable  $x$  along  $x$  axis and  $y$  along  $y$  axis. Now, Choose a line passing through the dots of scatter diagram. then that line becomes the line of best fit



If it satisfies the principle of least squares

Let  $y = f(x)$  be a bivariate data, then the principle of least squares states that the sum of the squares of distance b/w the given points and points in the line should be the least.



$\sum d_i^2$  is least  
line of best fit.

Fitting a straight line  $y = a + bx$

Let the straight line be  $y = a + bx$ .

Normal equations are

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

Solving the normal equations, we get the values of  $a$  and  $b$ . Substituting the values, we get the straight line  $y = a + bx$

using the principle of least square fit a straight line for following data

$x: 5 \quad 10 \quad 15 \quad 20 \quad 25$

$y: 15 \quad 19 \quad 23 \quad 26 \quad 30$

let the straight line be  $y = \frac{w}{n}a + b \cdot x$   
 $na + b \sum x$

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 5   | 15  | 25    | 225   | 75   |
| 10  | 19  | 100   | 361   | 190  |
| 15  | 23  | 225   | 529   | 345  |
| 20  | 26  | 400   | 676   | 520  |
| 25  | 30  | 625   | 900   | 750  |

$$\sum x = 75 \quad \sum y = 113 \quad \sum x^2 = 1375 \quad \sum y^2 = 2691$$

$$\sum xy = 1880$$

Normal equations are,

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$



$$113 = 5a + b \cdot 75$$

$$1880 = 75a + b \cdot 1375$$

$$5a + 75b = 113 \quad \rightarrow (1) \times 15$$

$$75a + 1375b = 1880 \quad \rightarrow (2)$$

$$~~75a + 1125b = 1695~~ -$$

$$~~75a + 1375b = 1880~~$$

$$-250b = -185$$

$$b = \underline{\underline{0.74}}$$

$$5a + (0.74 \times 75) = 113.$$

$$5a + 55.5 = 113.$$

$$5a = 57.5 \quad (57.5/5).$$

$$a = \underline{\underline{11.5}}$$

Straight line

$$y = \underline{\underline{11.5 + 0.74x}}$$

? fit a Straight line for the following data using normal equations method.

$x: 1 \ 3 \ 5 \ 7 \ 8 \ 10$

$y: 8 \ 12 \ 15 \ 17 \ 18 \ 20$

$$(y = a + bx)$$

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 1   | 8   | 1     | 64    | 8    |
| 3   | 12  | 9     | 144   | 36   |
| 5   | 15  | 25    | 225   | 75   |
| 7   | 17  | 49    | 289   | 119  |
| 8   | 18  | 64    | 324   | 144  |
| 10  | 20  | 100   | 400   | 200  |

$$\sum x = 34 \quad \sum y = 90 \quad \sum x^2 = 248$$

$$\sum y^2 = 1446$$

$$\sum xy = 582$$

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

$$90 = 6a + b \cdot 34$$

$$582 = a \cdot 34 + b \cdot 248$$

$$6a + 34b = 90$$

$$34a + 248b = 582$$



$$(1) \times 34 \Rightarrow 204a + 1156b = 3060$$

$$(2) \times 6 \Rightarrow 204a + 1488b = 3492$$

$$-332b = -432$$

$$b = 1.30$$

$$6a + 34b = 90$$

$$6a + 44.2 = 90$$

$$6a = 90 - 44.2$$

$$6a = 45.8$$

$$a = 7.63$$

$\therefore$  Straight line is  $y = 7.63 + 1.3x$

? fit a straight line of the form  $y = ax + b$  for the following data

|     |   |   |    |    |   |
|-----|---|---|----|----|---|
| $x$ | 1 | 2 | 3  | 4  | 5 |
| $y$ | 7 | 9 | 10 | 11 |   |

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 2   | 7   | 4     | 49    | 14   |
| 3   | 9   | 9     | 81    | 27   |
| 4   | 10  | 16    | 100   | 40   |
| 5   | 11  | 25    | 121   | 55   |

$$\sum x = 14 \quad \sum y = 37 \quad \sum x^2 = 54$$

$$\sum y^2 = 351 \quad \sum xy = 136$$

Normal equations are

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x$$

$$37 = 14a + 4b \quad \times 14$$

$$136 = 54a + 14b \quad \times 4$$

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$$196a + 56b = 518$$

$$216a + 56b = 544$$

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$$-20a = -26$$



$$a = 26/20 = 1.3$$

$$14a + 4b = 37$$

$$14 \times 1.3 + 4b = 37$$

$$4b = 37 - 18.2$$

$$4b = 18.8$$

$$b = 18.8/4 = 4.7$$

Straight line is  $1.3x + 4.7$

Fitting a parabola  $y = a + bx + cx^2$

Suppose we are given a bivariate data of  $x$  and  $y$  to fit a parabola, the normal eqn's are given by

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

Solving these normal equations we get the values of  $a$ ,  $b$  and  $c$ . Substituting  $a$ ,  $b$ ,  $c$  in the parabola, we fit the parabola for the given data.

Fit a parabola for the following data.

$x : 1 \quad 2 \quad 3 \quad 4 \quad 5$

$y : 3 \quad 9 \quad 16 \quad 25 \quad 40$

| $x$ | $y$ | $x^2$ | $x^3$ | $x^4$ | $xy$ | $x^2y$ |
|-----|-----|-------|-------|-------|------|--------|
| 1   | 3   | 1     | 1     | 1     | 3    | 3      |
| 2   | 9   | 4     | 8     | 16    | 18   | 36     |
| 3   | 16  | 9     | 27    | 81    | 48   | 144    |
| 4   | 25  | 16    | 64    | 256   | 100  | 400    |
| 5   | 40  | 25    | 125   | 625   | 200  | 1000   |
| 15  | 93  | 55    | 225   | 979   | 369  | 1583   |

Normal equations are :-



$$\sum y = na + b \sum x + c \sum x^2.$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3.$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4.$$

$$5a + 15b + 55c = 93 \quad \text{--- (1)}$$

$$15a + 55b + 225c = 369 \quad \text{--- (2)}$$

$$55a + 225b + 979c = 1583 \quad \text{--- (3)}$$

$$(1) \times 3$$

$$\begin{array}{r} 15a + 45b + 165c = 279 \\ 15a + 55b + 225c = 369 \\ \hline -10b + 60c = -90 \end{array}$$

$$10b + 60c = 90 \quad \text{--- (4)}$$

$$(2) \times 55$$

$$825a + 3025b + 12375c = 20295$$

$$(3) \times 15$$

$$825a + 3375b + 14685c = 23745$$

$$-350b - 2310c = -3450$$

$$350b + 2310c = 3450 \quad \text{--- (5)}$$

$$(A) \times 35 \rightarrow 3506 + 2100C = 3150$$

$$3506 + 2310C = 3450$$

$$-210C = -300$$

$$C = \frac{300}{210} = \underline{\underline{1.428}}$$

Substitute  $C = 1.428$  in (4)

$$10b + 60(1.428) = 90$$

$$10b = 90 - 85.68 = 4.32$$

$$b = \frac{4.32}{10} = \underline{\underline{0.432}}$$

$$5a + 15(0.432) + 55(1.428) = 93$$

$$5a = 93 - 6.48 - 78.54 = 7.98$$

$$a = \frac{7.98}{5} = \underline{\underline{1.596}}$$

∴ Parabola  $y = \underline{\underline{1.596 + 0.432x + 1.428x^2}}$

## Regression

When two variables are correlated, we can express one variable in terms of the other that means there is an equation connecting the two variables. This equation



tion is regression equation.

The line of best fit is also called regression line. There are 2 regression equations namely regression equation of  $y$  on  $x$  and  $x$  on  $y$ .

regression eq<sup>n</sup> of  $y$  on  $x$  is given by  
 $y - \bar{y} = b_{yx}(x - \bar{x})$  where  $b_{yx}$  is called  
regression coefficient on  
 $y$  on  $x$  given by  $b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$

$\bar{x}$  - A.M of  $x$

$\bar{y}$  - A.M of  $y$ .

regression eq<sup>n</sup> of  $x$  on  $y$  is

$x - \bar{x} = b_{xy}(y - \bar{y})$  where  $b_{xy}$  is

called regression coefficient of  $x$  on  $y$   
given by  $b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$

We have Correlation Coefficient

$$r = \sqrt{b_{yx} \times b_{xy}}$$

Ques:-

from the following data obtain the regression eq<sup>n</sup>s

x :- 6    2    10    4    8

y :- 9    11    5    8    7

| x  | y  | x <sup>2</sup> | y <sup>2</sup> | xy  |
|----|----|----------------|----------------|-----|
| 6  | 9  | 36             | 81             | 54  |
| 2  | 11 | 4              | 121            | 22  |
| 10 | 5  | 100            | 25             | 50  |
| 4  | 8  | 16             | 64             | 32  |
| 8  | 7  | 64             | 49             | 56  |
| 30 | 40 | 220            | 340            | 214 |

$$\bar{x} = \frac{\sum x}{n} = \frac{30}{5} = 6$$

$$\bar{y} = \frac{\sum y}{n} = \frac{40}{5} = 8$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

$$= \frac{5 \times 214 - 30 \times 40}{5 \times 340 - 40^2}$$



$$= \frac{-130}{100} = -1.3 //$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - \sum x^2}$$

$$= \frac{5 \times 214 - 30 \times 40}{1100 - 900} = \frac{-130}{200} = \underline{\underline{-0.65}}$$

Regression eq<sup>n</sup> on y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 8 = -1.3(x - 6)$$

$$= -1.3x + 7.8$$

$$y = -1.3x + 7.8 + 8$$

$$y = -1.3x + 15.8$$

Regression eq<sup>n</sup> on x on y is.

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 6 = -0.65(y - 8)$$

$$x - 6 = -0.65y + 5.2$$

$$x = -0.65y + 5.2 + 6$$

$$= -0.65y + 11.2$$


---

? The height and weight of a sample of 6 stickers Manufactured by a factory as given below. find the regression eqns and correlation Coefficient  $r$ .

height (x) :- 3    5    6    7    10    11  
 weight (y) :- 8    12    11    14    16    17

| x  | y  | $x^2$ | $y^2$ | $xy$ |
|----|----|-------|-------|------|
| 3  | 8  | 9     | 64    | 24   |
| 5  | 12 | 25    | 144   | 60   |
| 6  | 11 | 36    | 121   | 66   |
| 7  | 14 | 49    | 196   | 98   |
| 10 | 16 | 100   | 256   | 160  |
| 11 | 17 | 121   | 289   | 187  |
| 42 | 78 | 340   | 1070  | 595  |



$$\bar{x} = \frac{42}{6} = 7$$

$$\bar{y} = \frac{78}{6} = 13$$

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum y^2 - (\sum y)^2}$$

$$= \frac{6 \times 595 - 42 \cdot 78}{6 \times 1070 - (78)^2}$$

$$= \frac{3570 - 3276}{6420 - 6084}$$

$$= \frac{294}{336}$$

$$= \underline{\underline{0.875}}$$

$$b_{xy} = \frac{294}{6 \times 340 - (42)^2}$$

$$= \frac{294}{276} = \underline{\underline{1.065}}$$

Regression eq<sup>n</sup> on y on x is

$$y - 13 = 0.875(x - 7)$$

$$y - 13 = 0.875x - 6.125 + 13$$

$$y = \underline{\underline{0.875x + 6.875}}$$

regression eqn on  $x$  on  $y$

$$x - 7 = 1.065 (y - 13)$$

$$x = 1.065y - 13.845 + 7$$
$$= 1.065y - 6.845$$

$$r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{0.931} = \underline{\underline{0.964}}$$